

Supplementary Material for TokBench

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Overview

This supplementary material provides additional details and experimental results to complement the main paper. Section A covers dataset licenses, the complete composition and curation statistics of TokBench (including per-subset filtering funnels and an analysis of the consistency of OCR-based filtering), difficulty rating definitions, the face annotation protocol, the full user study protocol, and tokenizer model specifications. Section B presents multi-resolution evaluation results for text and face reconstruction, video evaluation at 480 resolution, temporal stability analysis of video reconstruction, analysis of the relationship between reconstruction and generation performance, and additional metric validation experiments. Section C describes ablation studies on the impact of training data composition. Section D provides qualitative reconstruction comparisons at 256 and 1024 resolutions.

A Dataset Details

A.1 Dataset Licenses

TokBench is curated from publicly available datasets for research purposes. We list the license information of all source datasets in Table 1. All datasets are used in compliance with their respective licenses and are intended solely for non-commercial academic research. We do not redistribute the original images or videos; our benchmark provides only annotations and evaluation scripts, with instructions to download source data from their official repositories.

A.2 Benchmark Composition and Curation

Overall composition. Table 2 summarizes the final composition of TokBench. The benchmark contains 12,398 images (6,000 text + 6,398 face) and 402 video clips (74 text + 328 face). Image subsets are quantified at the instance level (text or face instances), while video subsets are additionally reported by frames, reflecting the per-frame evaluation protocol.

Table 1: License information for source datasets used in TokBench.

Dataset	Type	License
<i>Image – Text</i>		
Total-Text	Scene text	BSD-3-Clause
TextOCR	Scene text	CC BY 4.0
ICDAR 2013	Scene text	Research only
ICDAR 2015	Scene text	Research only
SROIE	Receipt	MIT
CORD	Receipt	CC BY 4.0
DocVQA	Document	MIT
InfographicVQA	Document	MIT
<i>Image – Face</i>		
WFLW	Face	Research only [†]
<i>Video – Text</i>		
ICDAR 2015 Video	Video text	Research only
DSTextV2	Video text	Research only
<i>Video – Face</i>		
VideoMME	Video	Research only
MVBench	Video	CC BY 4.0 [‡]
MMBench-Video	Video	CC BY-NC-SA 4.0

[†] Images sourced from WIDER Face (non-commercial research only).

[‡] Contains videos from multiple sources with varying licenses.

Curation funnel. Table 3 details the curation funnel for each subset at the image/video level: the source datasets, the raw count before curation, the number removed, and the final retained count. The text-image and text-video pipelines filter at the *instance* level on top of the retained images/clips, and the corresponding instance-level statistics are described in the per-subset paragraphs below.

Image–Text subset. From the 22,855 raw images of the eight source datasets, we select 6,000 images balanced across sources and scales. The selected images contain 130,190 raw text instances, which are cleaned in two steps. First, extremely small instances are removed by the relative-scale criterion described in the main paper

Table 2: Composition of TokBench. Image subsets are evaluated at the instance level; video subsets are evaluated frame by frame.

Subset	Images / Clips	Frames	Instances
Image-Text	6,000 images	–	76,126
Image-Face	6,398 images	–	17,700
Video-Text	74 clips	15,921	347,468
Video-Face	328 clips	25,980	81,556
Total	12,398 images / 402 clips	41,901	522,850

Table 3: Curation funnel of TokBench at the image/video level. For Video-Face, the 125 retained source videos are cut into 328 three-second evaluation clips.

Subset	Sources	Raw	Removed	Final
Image-Text	IC13, IC15, Total-Text, TextOCR, CORD, SROIE, DocVQA, InfographicVQA	22,855	16,855	6,000
Image-Face	WFLW	6,551	153	6,398
Video-Text	IC13-15 Text-in-Video, DStextV2 (test)	75	1	74
Video-Face	VideoMME, MVBench, MMBench-Video	5,141	5,016	125 [†]

[†] Source videos; cut into 328 evaluation clips (25,980 frames).

($\min(h^t, w^t) < 5$ or $r^t < 0.005$), removing 24,654 instances and retaining 105,536 (a removal rate of 18.94%). Second, we keep only instances that the OCR model recognizes correctly on the *original* image, so that any score drop during evaluation is attributable to reconstruction rather than the recognizer’s inherent ceiling. This yields the final 76,126 valid text instances.

Consistency of OCR-based filtering. A natural concern is that the second cleaning step may inherit the bias of a specific OCR model. To verify its robustness, we repeat the OCR-based filtering with three recognizers (PARSeq, MASTER, and ViTSTR) and report the number of filtered instances and the filtering rate in Table 4. The three models remove a similar number of instances at similar rates. We further compute the pairwise and three-way Jaccard indices between the retained subsets (Table 5): the pairwise Jaccard index between any two subsets exceeds 0.9 at the instance level and 0.97 at the image level, indicating that different OCR models retain largely the same text instances. Together with Fig. 6 of the main paper, which shows that tokenizer rankings are stable across OCR backbones, these results indicate that neither the filtering procedure nor the evaluation conclusions are strongly biased toward a specific OCR model.

Table 4: OCR-based filtering statistics across three recognizers, applied to the 105,536 instances retained after scale-based cleaning.

OCR Model	PARSeq	MASTER	ViTSTR
Filtered instances	29,410	30,187	32,087
Filtering rate	27.87%	28.60%	30.40%

Characterization of filtered text instances. The filtered instances mainly include: (1) instances containing characters outside the OCR

Table 5: Jaccard indices between the subsets retained by different OCR models, at both the image and instance level.

Subsets	PARSeq-MASTER	PARSeq-ViTSTR	MASTER-ViTSTR	Three-way
Image level	0.9820	0.9737	0.9762	0.9661
Instance level	0.9139	0.9096	0.9055	0.8672

vocabulary, (2) instances consisting only of special symbols, (3) extremely small instances that are likely to vanish after resizing and reconstruction, and (4) difficult instances that cannot be correctly recognized even in the original image. The common issue of these samples is that they introduce unreliable evaluation signals, making it difficult to attribute performance degradation specifically to the reconstruction model. We note that hard fonts, blurred text, low-contrast text, and curved text are *not* explicitly excluded by our filtering strategy: such instances are retained whenever the OCR model recognizes them correctly on the original image, which preserves the diversity and difficulty of the evaluation data. Figure 1 visualizes representative retained and filtered samples.

Image-Face subset. Starting from the 6,551 images of WFLW, we remove 153 images with aspect ratios exceeding 2, retaining 6,398 images with 9,739 annotated faces. Since many images contain unannotated faces, we perform additional face detection on the original images with the Antelopev2 model from InsightFace, keep detections with confidence above 0.5 and relative scale above 0.05, and manually inspect every detected box (see Sec. A.4). This yields 17,700 valid evaluation faces in total.

Video-Text subset. The source datasets (ICDAR 2013–15 Text-in-Video and the DStextV2 test set) are already curated word-level video-text datasets, so only a single lightweight filtering pass is applied—identical in spirit to the image-text pipeline—removing instances that are too small in scale or not recognizable by the OCR model on the original frame. One of the 75 source videos is removed in this process, yielding 74 clips, 15,921 frames, and 347,468 valid text instances.

Video-Face subset. We build the face-video subset from three general-purpose video-understanding benchmarks—VideoMME, MVBench, and MMBench-Video—which together provide a raw pool of 5,141 videos. We first run face detection on one frame sampled per second and keep videos containing faces whose longer edge exceeds 512 pixels, yielding 2,480 face-containing videos. We then require a segment with continuous face presence for at least 3 seconds and more than 3 faces per frame, discarding videos in which faces appear only momentarily or that are dominated by single faces, which leaves 2,096 candidate videos. From this pool—aided by an additional pass profiling the maximum number of faces per video—we manually select 125 source videos that are face-centric with suitable image quality and scene diversity. Each selected video is cut into several 3-second clips, and every frame is processed with the Antelopev2 model to detect faces and record bounding boxes and 5-point landmarks, retaining detections with confidence above 0.5 and relative scale above 0.03. The resulting subset comprises 328 clips, 25,980 frames, and 81,556 face annotations.

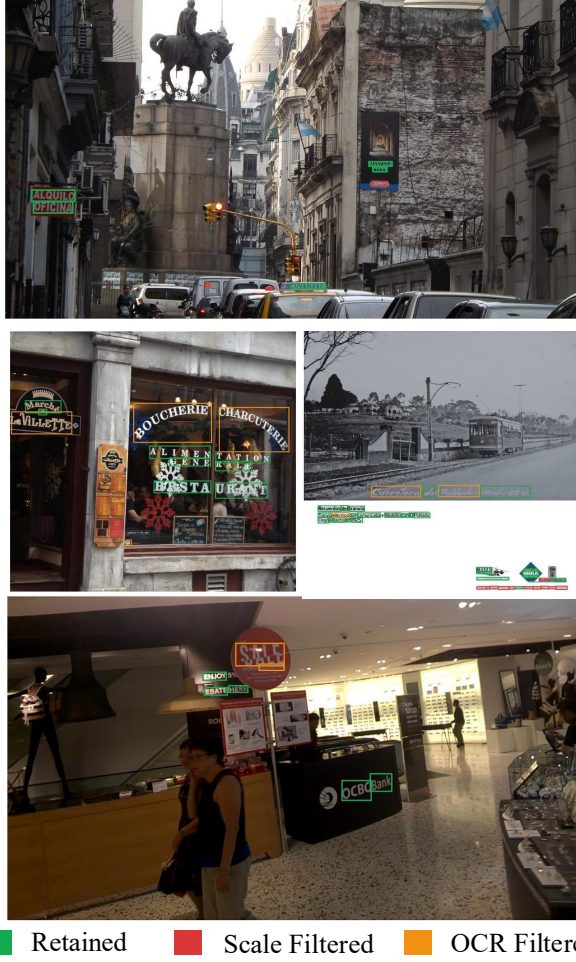


Figure 1: Examples of retained (green) and filtered (red and orange) text instances. Filtered samples are dominated by out-of-vocabulary characters, sub-5-pixel text, and text that is unrecognizable already in the original image.

A.3 Difficulty Rating

We classify target instances into three difficulty levels (Small, Medium, Large) based on their relative scale, as detailed in Table 6. The lower bound for each resolution is determined by the minimum pixel requirement for recognizable content.

For text, we assume at least 5×5 pixels are needed to represent a single character. At resolution L , instances with character-level scale below $5/L$ are excluded from evaluation. For example, at 256 resolution, the minimum character scale is $5/256 \approx 0.02$. For faces, we empirically set 25 pixels as the minimum for recognizable identity based on visualization and analysis of the resize upper bound, yielding a minimum face scale of $25/256 \approx 0.10$ at 256 resolution.

For video evaluation, since most videos follow a 16:9 aspect ratio and we resize the shorter edge to the target resolution (resulting in longer edges around 500 pixels), we adopt slightly more lenient

scale thresholds compared to image evaluation to accommodate these aspect ratio differences.

Table 6: Difficulty rating scale ranges. The lower bound for each resolution is determined by the minimum pixel requirement for recognizable faces. Instances below the Small threshold are excluded from evaluation at that resolution.

Type	Cat.	Res.	Small	Medium	Large
Image	Text	256	0.02 ~ 0.03	0.03 ~ 0.04	0.04 ~ 1.00
		512	0.01 ~ 0.02	0.02 ~ 0.03	0.03 ~ 1.00
		1024	0.005 ~ 0.01	0.01 ~ 0.02	0.02 ~ 1.00
	Face	256	0.10 ~ 0.20	0.20 ~ 0.30	0.30 ~ 1.00
		512	0.05 ~ 0.10	0.10 ~ 0.20	0.20 ~ 1.00
		1024	0.02 ~ 0.05	0.05 ~ 0.10	0.10 ~ 1.00
Video	Text	256	0.01 ~ 0.02	0.02 ~ 0.03	0.03 ~ 1.00
		480	0.005 ~ 0.01	0.01 ~ 0.02	0.02 ~ 1.00
	Face	256	0.05 ~ 0.10	0.10 ~ 0.20	0.20 ~ 1.00
		480	0.02 ~ 0.05	0.05 ~ 0.10	0.10 ~ 1.00

A.4 Face Annotation Protocol

We detail how the supplementary face annotations are obtained and why the evaluation is robust to the choice of detector and recognizer.

Keypoints for supplementarily detected faces. The additional faces beyond the WFLW ground-truth annotations are detected on the *original* images by the Antelopev2 model, which also outputs 5-point alignment keypoints. We keep only detections with confidence above 0.5 and relative scale above 0.05, and then manually inspect every detected box: any box not actually on a face, or on a face whose pose or clarity makes the identity indeterminate, is removed. The 17,700 retained faces therefore pass both the automatic thresholds and human verification, and our inspection confirms that the detected boxes and keypoints are of quality comparable to the WFLW ground-truth annotations.

Independence from detector behavior. The detector is applied only to the clean original image and never to any reconstruction. The verified original-image keypoints are applied unchanged to align and crop the *same* region from both the original and the reconstructed image, and F-Sim is the cosine similarity of the two resulting embeddings. The keypoints thus define a fixed reference geometry shared by both crops: any idiosyncrasy of the detector is applied identically to both sides and cancels in the comparison, so it cannot favor or penalize any tokenizer. F-Sim measures reconstruction-induced identity drift against a fixed reference rather than absolute recognition accuracy, and this holds regardless of the detector or recognizer family. Consistently, Fig. 6 of the main paper shows that tokenizer rankings are unchanged across recognition backbones (Antelope-V2, Buffalo-L, Buffalo-S).

A.5 User Study Protocol

We provide the complete protocol of the user study reported in Sec. 3.4 of the main paper.

Construction and sampling. Each triplet pairs one original image from the TokBench evaluation set with two reconstructions of it produced by two different tokenizers. For each original image, the two tokenizers are drawn at random from the full benchmarked set, and one resolution is drawn from {256, 512, 1024}. Triplets are sampled from a large candidate pool (~1,000), and each retained triplet is rated by multiple independent participants to form a per-triplet majority vote. The prompt shown to participants is: “Referencing the original image, select the reconstruction with higher quality.”

Balance and coverage. Triplets are drawn in roughly equal proportion from the text and face subsets, and—because tokenizers and resolutions are sampled at random—span all benchmarked tokenizers and all three resolutions rather than any fixed subset.

Randomization. The two reconstructions are shown in a randomized A/B order, independently per item, removing position bias.

Quality control. Quality control is applied at two stages. (1) *Construction:* every triplet is manually inspected, and trivial cases—where one reconstruction is so visibly degraded that the choice requires no judgment—are discarded, so that the retained triplets constitute genuinely meaningful comparisons. (2) *Post-collection:* we screen participants by response time (mean 9 s per triplet) and remove 5 questionnaires averaging under 3 s to exclude careless answering. The reported 55 participants and 1,100 comparisons are the retained set (60 questionnaires collected, 5 removed). Agreement is robust across raters: every retained participant agrees with the metric on a majority of their own judgments.

From metrics to preference. We do not fuse the metrics. Each triplet is judged against the single metric of its domain—F-Sim for face triplets and T-ACC for text triplets—and we test whether the human-majority choice matches the metric-preferred reconstruction. Over 1,100 comparisons, the agreement is 87.0% with a Spearman rank correlation of 0.74 ($p = 2.0 \times 10^{-55}$).

A.6 Tokenizer Selection

We detail the specific model versions used in our evaluation. For continuous VAEs, we employ VA-VAE [25], SDXL [16], SD3.5 [4], and FLUX [8] VAEs obtained from their official HuggingFace repositories, along with the 32× downsampling DC-AE with 32 latent dimensions from SANA [23].

For discrete tokenizers, we adopt the ImageNet-trained VQ-GAN [5] (F16, codebook size 16,384) as the baseline. LlamaGen [18] is evaluated in both F16 and F8 variants. VAR [19] uses the largest VAR-d36 model (2.3B parameters). TiTok [26] uses the TiTok-L-32 configuration, representing each image with 32 tokens. For Open-MAGVIT2 [11], we evaluate both F16 and F8 models trained on ImageNet, as well as an F16 model pretrained on 100M images with a codebook size of 262,144. MaskBit [22] uses the 12-bit variant. OmniTokenizer [21] uses the recommended imagenet_k600 version. FlexTok [1] uses the version trained on the DFN dataset [6]. UniTok [12] uses the released checkpoint with 4 sub-codebooks.

B Detailed Evaluation Results

B.1 Multi-Resolution Text Evaluation

Table 7 presents text reconstruction results across three resolutions (256, 512, 1024). Most tokenizers show progressive improvement with increasing resolution, as higher resolutions preserve more pixel-level detail for text characters. However, small-scale text remains challenging across all resolutions: even at 1024, discrete tokenizers with F16 downsampling achieve less than 5% T-ACC on small-scale text. The performance gap between continuous and discrete tokenizers widens significantly at higher resolutions. At 1024 resolution, FLUX’s VAE achieves T-NED comparable to the resize upper bound, while most discrete tokenizers still lag considerably. Notably, since many original text images exceed 1024 pixels, the resize baseline itself cannot achieve perfect T-ACC at this resolution. Furthermore, discrepancies between conventional metrics and T-ACC become more pronounced at higher resolutions: for instance, at 512 resolution LlamaGen and TokenFlow show similar LPIPS scores but differ substantially in T-ACC, and at 1024 resolution UniTok and Open-MAGVIT2 exhibit similar rFID yet diverge significantly in text reconstruction quality.

B.2 Multi-Resolution Face Evaluation

Table 8 presents face reconstruction results across three resolutions. Since most face images do not exceed 1024 pixels, the resize baseline at 1024 resolution preserves nearly identical facial details, yielding F-Sim close to 1.0. At this resolution, continuous VAEs from SD3.5 and FLUX achieve near-perfect performance, while even the best discrete tokenizers only reach approximately 0.5 F-Sim for small-scale faces. This highlights a persistent gap between discrete and continuous methods for fine-grained facial identity preservation at small scales. At lower resolutions, performance degrades substantially for all methods, with small-scale faces proving particularly challenging: at 256 resolution, most F16 discrete tokenizers achieve F-Sim below 0.15 on small-scale faces. Compared to text reconstruction, face reconstruction generally approaches the resize upper bound more closely, likely because existing tokenizers have been exposed to extensive facial data during training.

B.3 Video Evaluation at 480 Resolution

Table 9 presents video reconstruction results at 480 resolution, complementing the 256-resolution results in the main paper. At this higher resolution, all tokenizers show improved performance, with continuous VAEs benefiting more substantially. Hunyuan-Video and CogVideoX VAEs achieve strong text and face reconstruction at this resolution, narrowing the gap with the resize upper bound. The discrete Cosmos-VAE variants, however, continue to underperform significantly, particularly on small-scale text where T-ACC remains below 5% even at 480 resolution. Comparing across resolutions, the relative ranking of tokenizers remains largely consistent, confirming that TokBench provides stable and reliable evaluation across different resolution settings.

B.4 Temporal Stability of Video Reconstruction

Beyond per-frame averages, we quantify how strongly reconstruction quality fluctuates over time. For each clip, we first compute a

Table 7: Performance of discrete and continuous tokenizer on TokBench text-set.

Type	Method	Factor	T-ACC(%) $\uparrow$				T-NED(%) $\uparrow$				rFID $\downarrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$
			Small	Medium	Large	Mean	Small	Medium	Large	Mean				
Resolution: 256 $\times$ 256														
Discrete	Resize	1 $\times$	86.05	94.65	98.37	93.02	92.98	97.22	99.38	96.53	5.66	0.07	25.40	0.81
	TiTok	1D	0.05	0.06	0.17	0.09	3.04	4.07	5.58	4.23	18.41	0.50	13.80	0.50
	FlexTok	1D	0.55	2.24	18.06	6.95	7.80	14.26	41.21	21.09	11.01	0.31	17.58	0.61
	VQGAN	16 $\times$	0.05	0.12	3.14	1.10	4.34	5.33	15.00	8.22	15.66	0.33	17.17	0.58
	Chameleon	16 $\times$	0.11	0.31	8.19	2.87	4.67	6.65	24.91	12.08	17.60	0.33	17.66	0.59
	LlamaGen	16 $\times$	0.16	0.44	12.25	4.28	5.41	7.50	31.40	14.77	14.23	0.29	18.04	0.61
	VAR	16 $\times$	1.24	6.72	39.26	15.74	10.89	26.26	65.42	34.19	10.30	0.22	19.74	0.66
	MaskBit	16 $\times$	0.16	0.19	7.26	2.54	4.45	5.72	22.37	10.85	17.05	0.37	17.90	0.60
	TokenFlow	16 $\times$	0.28	1.62	18.29	6.73	6.41	12.34	42.64	20.46	11.04	0.26	18.61	0.62
	O-MAGVIT2	16 $\times$	0.34	1.49	20.73	7.52	6.46	12.41	44.10	20.99	10.18	0.25	18.85	0.63
	O-MAGVIT2(pretrain)	16 $\times$	0.80	3.17	27.76	10.58	9.59	19.15	54.02	27.59	9.83	0.24	19.15	0.64
	UniTok	16 $\times$	13.53	42.87	77.35	44.59	38.73	68.51	90.27	65.84	9.21	0.19	20.58	0.68
	OmniTokenizer	8 $\times$	2.14	8.46	51.28	20.63	13.24	30.50	73.67	39.14	12.70	0.30	14.73	0.62
	LlamaGen	8 $\times$	4.39	17.86	65.97	29.41	19.69	44.56	82.76	49.00	10.51	0.18	20.85	0.68
O-MAGVIT2	8 $\times$	9.33	34.16	77.24	40.24	30.82	59.89	89.19	59.97	8.99	0.16	21.71	0.71	
Continuous	DC-AE	32 $\times$	1.42	5.16	42.45	16.35	10.95	24.06	66.45	33.82	14.61	0.22	20.42	0.67
	VA-VAE	16 $\times$	6.92	28.25	75.96	37.04	25.14	55.30	88.52	56.32	8.22	0.16	21.94	0.71
	SD-XL	8 $\times$	6.94	24.83	70.85	34.21	25.03	50.96	85.03	53.68	8.93	0.18	21.69	0.70
	SD-3.5	8 $\times$	36.26	72.18	92.68	67.04	59.04	85.64	97.06	80.58	8.40	0.13	23.46	0.75
	FLUX.1-dev	8 $\times$	50.69	82.14	94.89	75.91	70.70	90.67	97.90	86.42	7.19	0.12	23.93	0.76
Resolution: 512 $\times$ 512														
Discrete	Resize	1 $\times$	92.51	98.18	98.86	96.52	96.25	99.24	99.64	98.38	0.26	0.01	29.80	0.91
	VQGAN	16 $\times$	0.15	0.76	17.45	6.12	5.20	8.99	37.77	17.32	6.87	0.19	19.24	0.65
	Chameleon	16 $\times$	0.60	2.67	31.39	11.55	7.63	17.82	54.95	26.80	5.61	0.17	19.81	0.66
	LlamaGen	16 $\times$	0.67	3.93	40.43	15.01	7.76	20.17	63.39	30.44	5.28	0.15	20.21	0.68
	VAR	16 $\times$	3.71	20.59	63.62	29.31	18.01	49.56	82.44	50.00	3.78	0.12	21.27	0.73
	TokenFlow	16 $\times$	1.06	6.27	44.88	17.40	10.00	28.39	68.07	35.49	5.14	0.15	20.46	0.68
	O-MAGVIT2	16 $\times$	1.40	9.51	54.04	21.65	10.79	33.15	74.94	39.63	3.65	0.13	21.11	0.71
	O-MAGVIT2(pretrain)	16 $\times$	3.02	16.25	62.72	27.33	16.87	44.50	80.48	47.28	3.51	0.12	21.54	0.72
	UniTok	16 $\times$	17.25	51.86	81.20	50.10	44.75	76.28	92.20	71.08	3.27	0.09	22.54	0.76
	OmniTokenizer	8 $\times$	6.21	38.33	82.91	42.48	23.44	65.99	92.66	60.70	5.67	0.20	15.00	0.67
	LlamaGen	8 $\times$	12.13	56.66	88.89	52.56	34.11	77.45	95.39	68.99	2.60	0.07	23.46	0.78
O-MAGVIT2	8 $\times$	20.66	70.36	90.66	60.56	46.60	85.41	96.00	76.00	2.34	0.07	24.39	0.80	
Continuous	DC-AE	32 $\times$	5.31	30.10	79.33	38.25	20.91	57.78	89.98	56.22	2.33	0.09	23.24	0.76
	VA-VAE	16 $\times$	12.72	58.73	88.43	53.30	34.80	78.86	95.30	69.65	2.23	0.07	24.07	0.79
	SD-XL	8 $\times$	16.53	62.87	91.20	56.86	40.43	80.83	96.40	72.55	2.00	0.06	24.67	0.80
	SD-3.5	8 $\times$	56.55	91.64	97.33	81.84	75.56	96.44	98.91	90.30	1.33	0.03	26.57	0.85
	FLUX.1-dev	8 $\times$	70.29	94.62	98.02	87.64	84.67	97.65	99.26	93.86	0.73	0.03	27.25	0.86
Resolution: 1024 $\times$ 1024														
Discrete	Resize	1 $\times$	95.15	98.39	99.30	97.61	97.97	99.33	99.77	99.02	0.18	0.01	inf	0.96
	VQGAN	16 $\times$	0.76	2.69	41.53	15.00	7.90	15.03	63.47	28.80	4.03	0.11	21.67	0.74
	Chameleon	16 $\times$	3.00	8.22	59.33	23.52	14.46	29.14	77.56	40.39	2.98	0.09	22.33	0.75
	LlamaGen	16 $\times$	3.30	10.62	67.63	27.19	14.13	33.02	83.57	43.58	3.35	0.09	22.74	0.77
	VAR	16 $\times$	9.64	30.08	75.35	38.36	29.07	59.16	89.51	59.25	4.85	0.10	22.40	0.79
	TokenFlow	16 $\times$	4.46	14.86	68.57	29.30	18.62	41.21	84.43	48.09	3.34	0.09	23.26	0.78
	O-MAGVIT2	16 $\times$	5.76	20.74	77.71	34.74	19.42	47.04	89.63	52.03	2.47	0.07	23.86	0.80
	O-MAGVIT2(pretrain)	16 $\times$	9.08	29.35	79.77	39.40	28.65	57.43	90.78	58.95	2.32	0.07	24.46	0.81
	UniTok	16 $\times$	26.90	47.91	74.29	49.70	54.36	72.48	87.93	71.59	4.02	0.07	24.22	0.83
	OmniTokenizer	8 $\times$	14.27	54.67	91.49	53.48	36.63	76.92	96.53	70.02	4.13	0.16	15.30	0.74
	LlamaGen	8 $\times$	25.42	71.63	94.61	63.89	50.33	86.45	97.95	78.24	1.74	0.04	26.57	0.86
	O-MAGVIT2	8 $\times$	35.29	78.91	94.84	69.68	60.97	90.36	98.03	83.12	2.21	0.06	27.07	0.88
	Continuous	DC-AE	32 $\times$	15.32	48.36	92.72	52.14	35.47	71.69	96.89	68.02	1.11	0.04	27.01
VA-VAE		16 $\times$	25.14	69.54	93.84	62.84	48.94	85.17	97.52	77.21	1.59	0.04	27.31	0.87
SD-XL		8 $\times$	31.41	75.83	96.29	67.84	56.60	88.34	98.52	81.15	1.01	0.03	28.60	0.88
SD-3.5		8 $\times$	74.88	95.76	98.50	89.71	87.57	98.15	99.44	95.05	0.54	0.02	29.80	0.92
FLUX.1-dev		8 $\times$	83.71	96.83	98.72	93.09	92.68	98.69	99.52	96.96	0.41	0.01	30.55	0.94

Table 8: Performance of discrete and continuous tokenizer on TokBench face-set.

Type	Method	Factor	Similarity↑				rFID↓	LPIPS↓	PSNR↑	SSIM↑
			Small	Medium	Large	Mean				
Resolution: 256 × 256										
Discrete	Resize	1×	0.85	0.97	0.98	0.93	7.83	0.05	29.83	0.87
	TiTok	1D	0.03	0.03	0.05	0.04	23.11	0.53	13.31	0.43
	FlexTok	1D	0.06	0.12	0.25	0.15	13.54	0.38	17.18	0.54
	VQGAN	16×	0.05	0.08	0.17	0.10	18.08	0.38	17.39	0.52
	Chameleon	16×	0.08	0.15	0.30	0.18	25.87	0.39	17.94	0.53
	LlamaGen	16×	0.07	0.11	0.26	0.15	15.30	0.32	18.38	0.55
	VAR	16×	0.10	0.20	0.41	0.23	13.11	0.25	20.20	0.61
	MaskBit	16×	0.06	0.09	0.19	0.11	15.92	0.39	18.23	0.55
	TokenFlow	16×	0.07	0.13	0.26	0.15	13.43	0.30	18.85	0.56
	O-MAGVIT2	16×	0.08	0.15	0.34	0.19	12.91	0.29	19.24	0.58
	O-MAGVIT2(pretrain)	16×	0.08	0.16	0.35	0.20	12.92	0.29	19.49	0.59
	UniTok	16×	0.15	0.32	0.58	0.35	11.25	0.21	21.66	0.65
	OmniTokenizer	8×	0.15	0.34	0.61	0.37	12.06	0.31	15.53	0.56
	LlamaGen	8×	0.17	0.38	0.66	0.40	12.01	0.20	22.09	0.66
	O-MAGVIT2	8×	0.23	0.48	0.74	0.48	11.47	0.18	23.27	0.69
Continuous	DC-AE	32×	0.10	0.21	0.45	0.26	17.58	0.25	21.30	0.62
	VA-VAE	16×	0.22	0.48	0.76	0.49	9.26	0.16	23.85	0.70
	SD-XL	8×	0.18	0.40	0.69	0.42	11.19	0.20	23.29	0.68
	SD-3.5	8×	0.43	0.76	0.92	0.70	9.91	0.13	26.20	0.75
	FLUX.1-dev	8×	0.52	0.83	0.95	0.76	9.32	0.11	26.94	0.78
Resolution: 512 × 512										
Discrete	Resize	1×	0.95	0.99	1.00	0.98	0.08	0.00	37.34	0.97
	VQGAN	16×	0.08	0.11	0.37	0.19	7.33	0.23	20.42	0.61
	Chameleon	16×	0.13	0.21	0.50	0.28	6.62	0.22	20.98	0.61
	LlamaGen	16×	0.11	0.17	0.48	0.25	5.28	0.18	21.41	0.65
	VAR	16×	0.14	0.24	0.57	0.32	4.59	0.15	22.16	0.69
	TokenFlow	16×	0.11	0.16	0.45	0.24	6.48	0.19	21.39	0.64
	O-MAGVIT2	16×	0.13	0.22	0.58	0.31	4.67	0.16	22.40	0.67
	O-MAGVIT2(pretrain)	16×	0.13	0.22	0.57	0.31	4.55	0.16	22.66	0.68
	UniTok	16×	0.22	0.36	0.74	0.44	3.95	0.11	24.34	0.74
	OmniTokenizer	8×	0.24	0.45	0.80	0.50	5.11	0.20	15.93	0.63
	LlamaGen	8×	0.28	0.49	0.83	0.53	2.73	0.08	25.49	0.77
	O-MAGVIT2	8×	0.35	0.58	0.88	0.61	2.80	0.07	26.81	0.80
Continuous	DC-AE	32×	0.16	0.29	0.71	0.39	2.81	0.11	25.08	0.73
	VA-VAE	16×	0.31	0.54	0.87	0.57	2.41	0.07	26.84	0.79
	SD-XL	8×	0.29	0.51	0.87	0.55	2.46	0.07	27.14	0.79
	SD-3.5	8×	0.61	0.84	0.98	0.81	1.20	0.03	30.06	0.87
	FLUX.1-dev	8×	0.71	0.89	0.98	0.86	0.71	0.02	31.06	0.90
Resolution: 1024 × 1024										
Discrete	Resize	1×	1.0	1.0	1.0	1.0	0.01	0.00	inf	1.00
	VQGAN	16×	0.10	0.19	0.47	0.25	4.27	0.13	23.98	0.72
	Chameleon	16×	0.19	0.30	0.58	0.36	3.63	0.11	24.54	0.73
	LlamaGen	16×	0.15	0.27	0.57	0.33	3.45	0.10	24.77	0.76
	VAR	16×	0.23	0.34	0.62	0.40	6.21	0.12	23.67	0.77
	TokenFlow	16×	0.14	0.26	0.55	0.31	4.63	0.10	25.00	0.76
	O-MAGVIT2	16×	0.20	0.34	0.66	0.40	3.62	0.09	26.01	0.79
	O-MAGVIT2(pretrain)	16×	0.21	0.34	0.65	0.40	3.48	0.09	26.12	0.79
	UniTok	16×	0.33	0.50	0.76	0.53	3.78	0.07	26.54	0.84
	OmniTokenizer	8×	0.40	0.60	0.82	0.61	4.63	0.15	16.00	0.71
	LlamaGen	8×	0.49	0.66	0.87	0.67	2.11	0.04	28.89	0.87
	O-MAGVIT2	8×	0.57	0.74	0.91	0.74	2.11	0.05	29.92	0.89
Continuous	DC-AE	32×	0.27	0.45	0.78	0.50	1.44	0.05	29.48	0.84
	VA-VAE	16×	0.49	0.68	0.89	0.69	2.38	0.04	30.69	0.88
	SD-XL	8×	0.50	0.69	0.91	0.70	1.25	0.03	31.39	0.89
	SD-3.5	8×	0.86	0.94	0.99	0.93	0.42	0.01	33.07	0.96
	FLUX.1-dev	8×	0.92	0.97	0.99	0.96	0.24	0.01	33.61	0.97

Table 9: Performance of video tokenizer on TokBench-Video. The resolution refers specifically to the shorter edge of the videos, while maintaining the original aspect ratio throughout. The categorization into small, medium, and large scales is dynamically adjusted based on resolution.

Type	Method	Factor	T-ACC(%)↑				T-NED(%)↑				F-Sim↑			
			Small	Medium	Large	Mean	Small	Medium	Large	Mean	Small	Medium	Large	Mean
Resolution: 256														
Discrete	Resize	1 × 1 × 1	76.09	92.14	96.18	88.14	85.77	95.79	98.32	93.29	0.81	0.91	0.97	0.90
	Cosmos-VAE [15]	4 × 8 × 8	1.49	22.82	66.12	30.14	7.76	44.61	77.15	43.18	0.29	0.52	0.76	0.52
	Cosmos-VAE [15]	8 × 16 × 16	0.02	0.38	2.79	1.06	0.84	3.14	12.95	5.64	0.10	0.13	0.25	0.16
Continuous	Cosmos-VAE [15]	4 × 8 × 8	5.80	52.09	78.34	45.41	15.80	68.06	85.63	56.50	0.47	0.72	0.89	0.69
	Hunyuan-Video [7]	4 × 8 × 8	26.85	69.12	87.47	61.15	45.55	80.54	93.12	73.07	0.60	0.80	0.92	0.77
	CogVideoX [24]	4 × 8 × 8	24.80	72.47	86.34	61.21	43.06	82.29	92.41	72.59	0.58	0.78	0.91	0.76
	Cosmos-VAE [15]	8 × 16 × 16	0.45	6.23	48.99	18.56	3.25	24.62	64.08	30.65	0.21	0.39	0.65	0.42
	Step-Video [13]	8 × 16 × 16	17.39	61.40	82.41	53.73	33.16	75.67	89.76	66.19	0.48	0.69	0.86	0.67
Resolution: 480														
Discrete	Resize	1 × 1 × 1	64.44	90.74	96.92	84.04	77.71	95.72	98.57	90.67	0.82	0.89	0.95	0.89
	Cosmos-VAE [15]	4 × 8 × 8	0.90	20.32	73.71	31.64	6.74	41.81	83.53	44.03	0.44	0.60	0.80	0.61
	Cosmos-VAE [15]	8 × 16 × 16	0.02	0.90	13.82	4.91	0.85	4.20	27.02	10.69	0.19	0.18	0.31	0.23
Continuous	Cosmos-VAE [15]	4 × 8 × 8	5.30	46.80	86.82	46.31	14.99	64.63	92.20	57.27	0.60	0.77	0.90	0.76
	Hunyuan-Video [7]	4 × 8 × 8	28.65	64.49	91.83	61.66	44.43	77.83	95.83	72.70	0.69	0.82	0.92	0.81
	CogVideoX [24]	4 × 8 × 8	28.02	65.41	91.71	61.71	43.47	78.24	95.60	72.43	0.67	0.80	0.91	0.79
	Cosmos-VAE [15]	8 × 16 × 16	0.36	9.40	61.81	23.86	3.20	22.70	73.76	33.22	0.34	0.47	0.71	0.51
	Step-Video [13]	8 × 16 × 16	20.27	54.18	87.14	53.86	35.43	71.39	93.04	66.62	0.60	0.73	0.86	0.73

Table 10: Temporal stability of video reconstruction at 256 resolution, measured as the standard deviation of per-frame scores within each clip, averaged over valid clips. Lower is better.

Type	Method	Factor	T-ACC Std↓				T-NED Std↓				F-Sim Std↓			
			Small	Medium	Large	Mean	Small	Medium	Large	Mean	Small	Medium	Large	Mean
Discrete	Cosmos-VAE	4×8×8	0.001	0.047	0.114	0.054	0.009	0.055	0.079	0.048	0.011	0.010	0.007	0.009
	Cosmos-VAE	8×16×16	0.000	0.005	0.012	0.006	0.001	0.009	0.027	0.012	0.009	0.011	0.017	0.012
Continuous	Cosmos-VAE	4×8×8	0.013	0.104	0.133	0.083	0.029	0.081	0.081	0.064	0.012	0.010	0.004	0.009
	Hunyuan-Video	4×8×8	0.057	0.127	0.096	0.093	0.064	0.075	0.046	0.062	0.010	0.006	0.001	0.006
	CogVideoX	4×8×8	0.051	0.124	0.102	0.092	0.062	0.076	0.054	0.064	0.010	0.007	0.002	0.006
	Cosmos-VAE	8×16×16	0.000	0.011	0.086	0.032	0.003	0.025	0.071	0.033	0.010	0.011	0.013	0.011
	Step-Video	8×16×16	0.033	0.111	0.122	0.089	0.049	0.077	0.062	0.063	0.012	0.010	0.005	0.009

per-frame score (the mean T-ACC, T-NED, or F-Sim over the valid instances in that frame), then take the standard deviation of these per-frame scores across the clip, and finally average over all valid clips. A smaller value indicates that reconstruction quality does not fluctuate strongly from frame to frame. Note that this metric serves as a supplementary indicator of quality fluctuation rather than a full measure of temporal coherence. Results at 256 resolution are reported in Table 10.

We observe that very low std values can arise for two distinct reasons: stable high-quality reconstruction, or uniformly failed reconstruction (e.g., the discrete Cosmos-VAE at 8×16×16, whose T-ACC is near zero on small text in every frame). Temporal stability should therefore be read jointly with the absolute per-frame metrics.

B.5 Reconstruction vs. Generation Performance

A natural question is whether better reconstruction quality translates to better generation performance. To investigate this, we collect reported reconstruction and class-conditional generation results on the ImageNet benchmark from original publications. Since different tokenizers are trained with diverse generators, datasets, and schedules, these generation results are not directly comparable across methods but provide an indicative reference. We summarize the results in Table 11.

We observe several key findings from this comparison:

General correlation with notable exceptions. Tokenizers with better TokBench scores generally achieve better ImageNet rFID, suggesting that fine-grained reconstruction quality on text and faces is indicative of overall reconstruction fidelity. However, generation performance (gFID) does not always align with reconstruction ranking. For example, VAR ranks fifth in reconstruction but third in generation, highlighting that generative modeling strategy (e.g.,

Table 11: Comparison of reconstruction quality (on TokBench and ImageNet) and ImageNet class-conditional generation performance. gFID results are cited from original publications using their respective generators. ‘-’ indicates unavailable results.

Method	TokBench Face		TokBench Text		ImageNet Rec.		ImageNet Gen.		Generator
	F-Sim _m ↑	Rank	T-ACC _m (%)↑	Rank	rFID↓	Rank	gFID↓	Rank	
VQGAN	0.07	9	1.10	9	7.94	10	15.78	8	MaskGIT
LlamaGen	0.10	8	4.28	8	2.19	8	3.06	7	LlamaGen-L
VAR	0.16	5	15.74	5	0.90	5	1.73	3	VAR-d30
TiTok	0.04	10	0.09	10	2.21	9	2.77	6	MaskGIT
TokenFlow	0.11	7	6.73	7	1.37	7	-	-	-
UniTok	0.23	4	44.59	1	0.41	3	2.51	5	LlamaGen-L
O-MAGVIT2	0.13	6	7.52	6	1.17	6	2.33	4	LlamaGen-L
O-MAGVIT2(F8)	0.32	1	40.24	2	0.34	2	-	-	-
VA-VAE	0.32	1	37.04	3	0.28	1	1.35	1	LlamaGen-XL
DC-AE	0.26	3	16.35	4	0.69	4	1.72	2	SiT-XL

next-scale prediction) can compensate for reconstruction limitations.

Reconstruction quality is necessary but not sufficient. TiTok achieves the lowest reconstruction quality on TokBench yet attains mid-ranked gFID, suggesting that generation focuses on plausible sampling rather than detail preservation. Conversely, O-MAGVIT2(F8) achieves excellent reconstruction but its longer token sequences make autoregressive generation computationally expensive, illustrating the trade-off between reconstruction fidelity and generation efficiency.

Our metrics provide complementary insights. While rFID captures global reconstruction quality, our T-ACC and F-Sim reveal fine-grained differences that rFID misses. For instance, UniTok and O-MAGVIT2 have comparable rFID (0.41 vs. 1.17) but differ dramatically in T-ACC (44.59% vs. 7.52%), indicating that UniTok’s multi-codebook quantization is substantially better at preserving text details. These distinctions are invisible to conventional metrics alone.

These observations confirm that while reconstruction quality provides a necessary foundation for generation fidelity, generation performance is additionally influenced by the generator architecture, training configuration, and the modeling difficulty of the latent space. TokBench serves as an efficient tool for evaluating the reconstruction aspect of this equation.

Controlled comparisons and confounding factors. The benchmarked tokenizers differ in training data, resolution, and training recipe, so cross-tokenizer comparisons reflect whole training setups rather than architecture alone. Where the benchmark allows cleaner attribution, we highlight controlled pairs: the Cosmos-VAE discrete-vs-continuous pair shares an identical architecture and isolates the effect of quantization, and the LlamaGen and Open-MAGVIT2 F16-vs-F8 pairs share identical architectures and isolate the effect of the spatial downsampling factor. For tokenizers compared under inconsistent training settings, conclusions about individual design choices should be drawn with caution.

B.6 Evaluation on Texture Reconstruction

To examine whether our benchmark’s findings generalize beyond text and faces, we conduct additional experiments on the Describable Textures Dataset (DTD), evaluating reconstruction performance using conventional metrics. As shown in Table 12, the overall ranking of tokenizers on textures is broadly consistent with our findings on faces and text, confirming the generalizability of our main conclusions. However, conventional metrics (rFID, LPIPS, PSNR, SSIM) yield inconsistent fine-grained rankings among top-performing models, highlighting the difficulty of reliable evaluation using generic metrics alone. This reinforces our motivation for focusing on text and faces, where task-aware metrics (OCR accuracy, face recognition similarity) provide objective and unambiguous evaluation. Moreover, text inherently shares critical properties with fine textures and thin lines: dense text blocks can be perceived as a form of fine texture, and characters are fundamentally composed of thin strokes whose accurate reconstruction is directly captured by our OCR-based metrics.

Table 12: Texture reconstruction evaluation on DTD. Rankings are shown after each value. The overall trend is consistent with TokBench results, but conventional metrics show inconsistent fine-grained rankings among top models.

Method	TokBench		DTD Texture				
	T-ACC _m (%)↑	F-Sim _m ↑	rFID↓	LPIPS↓	PSNR↑	SSIM↑	
VQGAN	1.10 / 14	0.07 / 14	15.51 / 14	0.37 / 14	18.55 / 14	0.35 / 14	
Chameleon	2.87 / 13	0.12 / 11	13.67 / 13	0.36 / 13	18.98 / 13	0.37 / 13	
LlamaGen	4.28 / 12	0.10 / 13	11.75 / 12	0.31 / 12	19.47 / 12	0.41 / 11	
VAR	15.74 / 8	0.16 / 8	9.03 / 8	0.25 / 8	21.25 / 8	0.50 / 8	
TiTok	0.09 / 15	0.04 / 15	19.85 / 15	0.51 / 15	15.43 / 15	0.25 / 15	
TokenFlow	6.73 / 11	0.11 / 12	11.03 / 11	0.29 / 11	19.87 / 11	0.41 / 11	
UniTok	44.59 / 3	0.23 / 7	8.41 / 6	0.21 / 6	22.62 / 7	0.56 / 7	
O-MAGVIT2	7.52 / 10	0.13 / 9	9.47 / 9	0.29 / 9	20.34 / 10	0.44 / 10	
O-MAGVIT2(pt)	10.58 / 9	0.13 / 9	9.59 / 10	0.29 / 9	20.46 / 9	0.45 / 9	
LlamaGen(F8)	29.41 / 7	0.26 / 6	7.61 / 5	0.19 / 4	22.70 / 6	0.57 / 5	
O-MAGVIT2(F8)	40.24 / 4	0.32 / 3	7.43 / 4	0.19 / 4	23.63 / 5	0.61 / 3	
VA-VAE	37.04 / 5	0.32 / 3	6.16 / 3	0.18 / 3	23.99 / 4	0.60 / 4	
SD-XL	34.21 / 6	0.27 / 5	8.58 / 7	0.21 / 6	23.79 / 3	0.57 / 5	
SD-3.5	67.04 / 2	0.49 / 2	5.82 / 2	0.14 / 2	26.24 / 2	0.70 / 2	
FLUX.1-dev	75.91 / 1	0.56 / 1	5.26 / 1	0.11 / 1	27.20 / 1	0.75 / 1	

B.7 T-ACC vs. Text Feature Similarity

An alternative to using OCR recognition accuracy for text evaluation is to compute feature similarity from the OCR encoder, analogous to F-Sim for faces. To compare these two approaches, we extract features from the PARSeq encoder and compute cosine similarity between original and reconstructed text regions. As shown in Table 13, both metrics yield nearly identical tokenizer rankings, with only minor differences at rank 10–11 (O-MAGVIT2 and TokenFlow). However, T-ACC provides substantially better discriminability: the similarity scores are compressed into a narrow range (0.50–0.92), making it difficult to distinguish between tokenizers with similar performance. In contrast, T-ACC spans a much wider range (0.09%–75.91%), offering clearer separation. Furthermore, recognition accuracy aligns more naturally with human judgment, which is typically based on whether text is readable rather than on abstract feature distances.

Table 13: Comparison of T-ACC and text feature similarity. Both metrics yield consistent rankings, but T-ACC offers better discriminability across the full performance range.

Method	T-ACC _m (%)↑	Rank	Similarity↑	Rank
VQGAN	1.10	14	0.60	14
Chameleon	2.87	13	0.63	13
LlamaGen	4.28	12	0.65	12
VAR	15.74	8	0.74	8
TiTok	0.09	15	0.50	15
TokenFlow	6.73	11	0.69	10
UniTok	44.59	3	0.84	3
O-MAGVIT2	7.52	10	0.68	11
O-MAGVIT2(pt)	10.58	9	0.72	9
LlamaGen(F8)	29.41	7	0.79	7
O-MAGVIT2(F8)	40.24	4	0.83	4
VA-VAE	37.04	5	0.82	6
SD-XL	34.21	6	0.82	5
SD-3.5	67.04	2	0.90	2
FLUX.1-dev	75.91	1	0.92	1

C Ablation Study

Table 14: Ablations on Training Data. While augmenting ImageNet with text-rich data yields performance improvements, the gains remain modest compared to architectural differences observed in Table 1 of the main paper.

Method	Data	T-ACC _s ↑	T-ACC _m ↑	T-NED _s ↑	T-NED _m ↑
F16	ImageNet	0.02	2.96	4.84	12.11
F16	ImageNet+Text	0.09	3.93	5.19	14.48
F8	ImageNet	2.99	25.99	16.25	45.09
F8	ImageNet+Text	3.42	27.51	18.05	47.36

C.1 Impact of Text-Rich Training Data

Since different tokenizers are typically trained on distinct datasets, a natural question is whether incorporating more text-rich data can improve text reconstruction quality. To investigate this, we conduct ablation studies following LlamaGen’s [18] training protocol. We

train both F16 and F8 VQGAN models on two dataset configurations: (1) ImageNet [3] alone, and (2) ImageNet augmented with an additional 230K text-rich images. Both configurations are evaluated on the TokBench text set, with results shown in Table 14.

The results demonstrate that incorporating text-rich training data improves T-ACC and T-NED scores across both downsampling factors. However, these improvements are relatively marginal compared to the gains achieved through architectural enhancements such as advanced quantization methods (e.g., MCQ in UniTok) or reduced downsampling ratios (F8 vs. F16). This suggests that while training data composition influences reconstruction quality, tokenizer architecture and quantization design remain the more critical factors for preserving fine-grained visual details.

C.2 Training Details

Our ablation uses the ImageNet [3] training set as the baseline. For the augmented configuration, we supplement with 230,000 text-rich images sourced from the training sets of Synth150K [9], ICDAR 2017 MLT [14], Total-Text [2], TextOCR [17], CTW1500 [10], and COCO-Text [20]. Importantly, the additional text images are drawn from different splits or entirely different datasets than those used in TokBench evaluation, ensuring no data leakage. Only the images are used for training; no text annotations are required. To ensure a fair comparison, both configurations are trained for 400,000 iterations under identical hyperparameters, following the default LlamaGen VQGAN training recipe.

D More Visualization

Tables 2 and 3 present qualitative comparisons of reconstruction results from different methods at 256 and 1024 resolutions respectively. At 256 resolution, most discrete tokenizers fail to accurately reconstruct text and faces, while the high-compression DC-AE also performs poorly. In contrast, SD3.5 and FLUX VAEs demonstrate significantly better visual quality. At 1024 resolution, both VAEs and low-compression (F8) discrete tokenizers achieve satisfactory results, though F8 Open-MAGVIT2 exhibits noticeable color distortion, and F16 discrete tokenizers still struggle with small-scale objects.

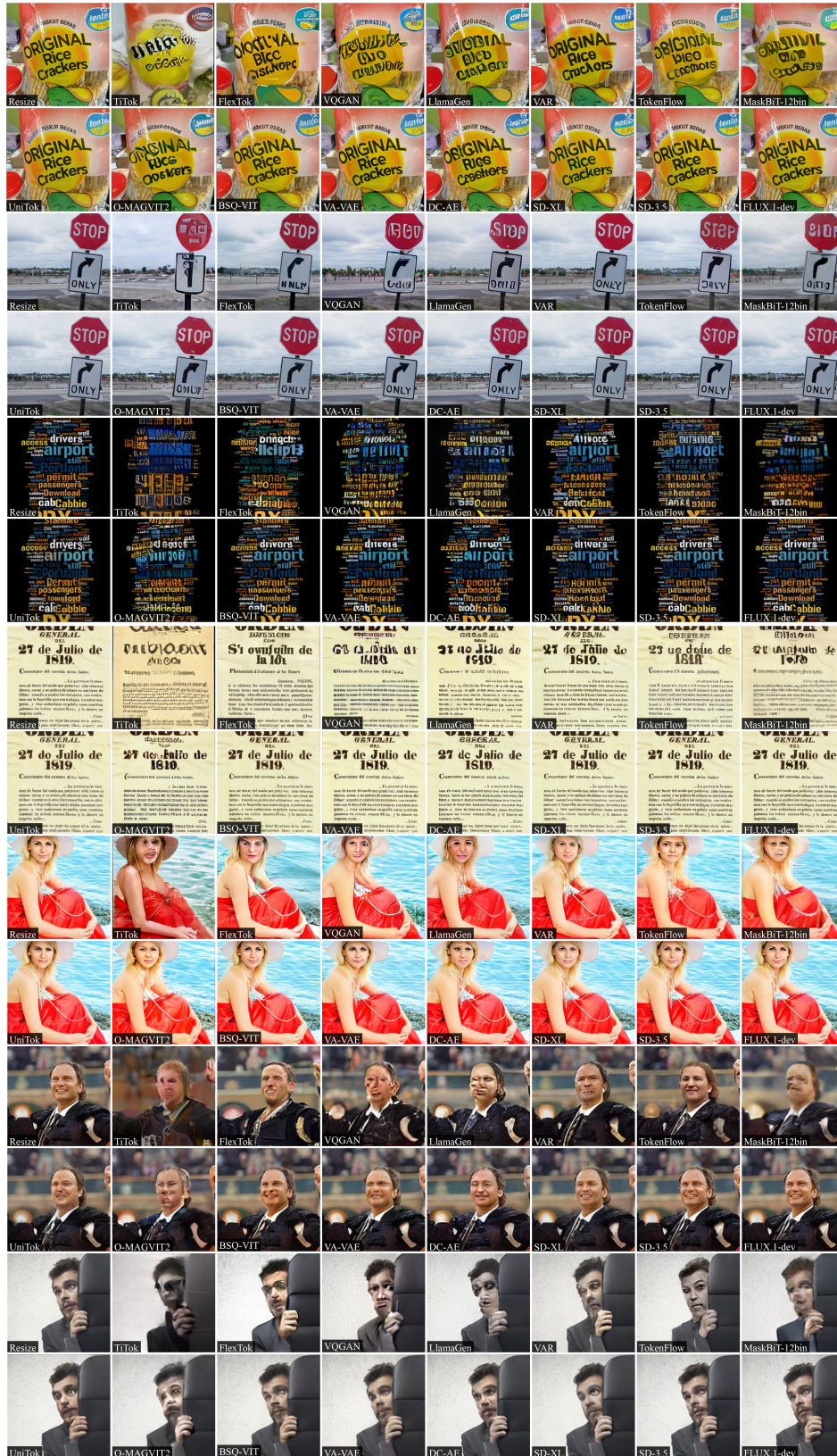
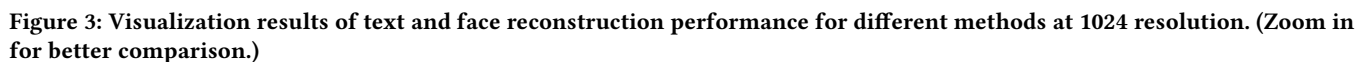


Figure 2: Visualization results of text and face reconstruction performance for different methods at 256 resolution. (Zoom in for better comparison.)



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